Quizrr Chapter-Wise Test for JEE Main - 2025

Q.1

The most abundant isotope of helium has a ${}_{2}^{4}\mathrm{H}$ nucleus whose mass is $6.6447 \times 10^{-27}\ \mathrm{kg}$. For this nucleus, find (a) the mass defect and (b) the binding energy.
Given: Mass of the electron: $m_{\mathrm{e}} = 5.485799 \times 10^{-4}\mathrm{u}$, mass of the proton: $m_p = 1.007276\mathrm{u}$ and mass of the neutron: $m_n = 1.008665\mathrm{u}$.

(A) 0.0503×10^{-27} kg, 28.3 Mev

(B) 0.03×10^{-27} kg, 28.3 Mev

(C) 0.0503×10^{-27} kg, 299.3 Mev

(D) 0.0803×10^{-27} kg, 288.3 Mev

Q.2

The nuclear radius of ${}_{8}\mathrm{O}^{16}$ is $3 \times 10^{-15}\ \mathrm{m}$. If an atomic mass unit is $1.67 \times 10^{-27}\ \mathrm{kg}$, then the nuclear density is approximately

(A) $2.35 \times 10^{17}\ \mathrm{g\ cm^{-3}}$

(B) $2.35 \times 10^{17}\ \mathrm{kg\ m^{-3}}$

(C) $2.35 \times 10^{17}\mathrm{gm^{-3}}$

(D) $2.35 \times 10^{17}\mathrm{kgmm^{-3}}$

Q.3

If the nuclear radius of ${}^{27}\mathrm{Al}$ is 3.6 Fermi, the approximate nuclear radius of ${}^{64}\mathrm{Cu}$ in Fermi is

(A) 4.8

(B) 3.6

(C) 2.4

(D) 1.2

Q.4

The nuclear radius of a nucleus with nucleon number 16 is $3 \times 10^{-15}\ \mathrm{m}$. Then, the nuclear radius of a nucleus with nucleon number 128 is

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(A) 3×10^{-15} m

(B) 1.5×10^{-15} m

(C) 6×10^{-15} m

(D) 4.5×10^{-15} m

Q.5

Atomic weight of boron is 10.81 and it has two isotopes ${}_5\mathrm{B}^{10}$ and ${}_5\mathrm{B}^{11}$. Then ratio of ${}_5\mathrm{B}^{10} : {}_5\mathrm{B}^{1}$. in nature would be

(A) $19 : 81$

(B) $10 : 11$

(C) $15 : 16$

(D) $81 : 19$

Q.6

If the nucleus ${}^{27}_{13}Al$ has nuclear radius of about $3.6\mathrm{fm}$, then ${}^{125}_{32}\mathrm{Te}$ would have its radius approximately as

(A) $9.6\mathrm{fm}$

(B) $12.0\mathrm{fm}$

(C) $4.8\mathrm{fm}$

(D) $6.0\mathrm{fm}$.

Q.7

The radius of germanium (Ge) nuclide is measured to be twice the radius of ${}^{9}_{4}\mathrm{Be}$. The number of nucleons in Ge are

(A) 74

(B) 75

(C) 72

(D) 73

Q.8

A nucleus disintegrates into two nuclear parts which have their velocities in the ratio $2 : 1$. Ratio of their nuclear sizes will be

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(A) $2^{1/3} : 1$

(B) $1 : 3^{1/2}$

(C) $3^{1/2} : 1$

(D) $1 : 2^{1/3}$

Q.9

If radius of the $^{27}_{12}\mathrm{Al}$ nucleus is taken to be $\mathrm{R_{Al}}$, then the radius of $^{125}_{53}$ Te nucleus is nearly:

(A) $\frac{5}{3}\mathrm{R_{Al}}$

(B) $\frac{3}{5}R_{\mathrm{Al}}$

(C) $\left(\frac{13}{53}\right)^{1/3}\mathrm{R_{Al}}$

(D) $\left(\frac{53}{13}\right)^{1/3}\mathrm{R_{AI}}$

Q.10

Calculate binding energy of $_{92}\mathrm{U}^{238}$.
Given $M\left(\mathrm{U}^{238}\right) = 238.050783$ amu, $m_n = 1.008665$ amu and $m_P = 1.007825\mathrm{amu}$

(A) $801.7\mathrm{MeV}$

(B) $18.7\mathrm{MeV}$

(C) $0.7\mathrm{MeV}$

(D) $1801.7\mathrm{MeV}$

Q.11

The binding energy of deuteron $\left(\begin{array}{c}2\\1\end{array}\right]$) is 1.15 MeV per nucleon and an alpha particle $\left(^4_2\mathrm{He}\right)$ has a binding energy of $7.1\mathrm{MeV}$ per nucleon. Then in the reaction
$^2_1\mathrm{H} + {}^2_1\mathrm{H} \longrightarrow {}^2_2\mathrm{He} + Q$
the energy released Q is:

(A) $5.95\mathrm{MeV}$

(B) $26.1\mathrm{MeV}$

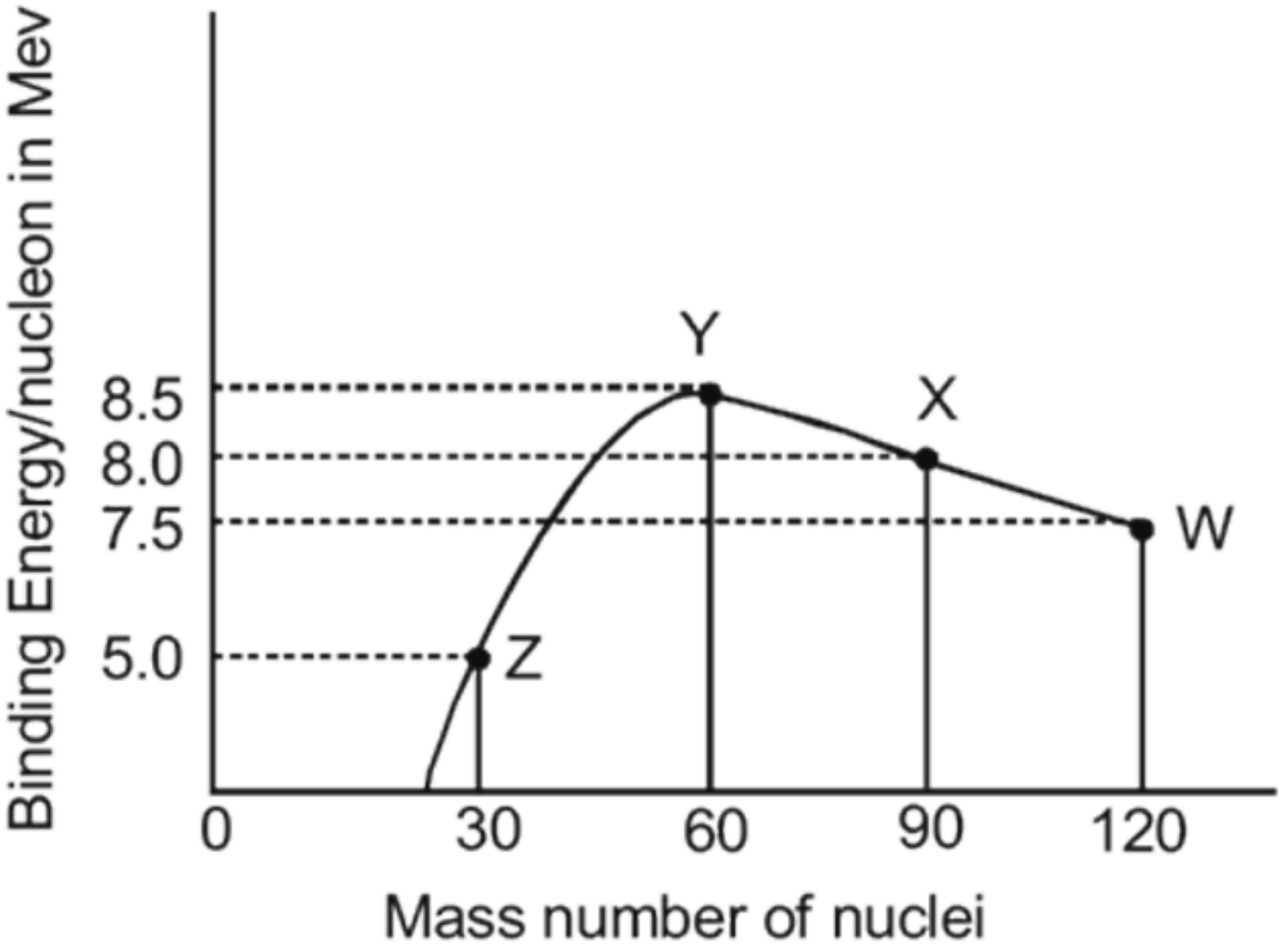
(C) $23.8\mathrm{MeV}$

(D) $289.4\mathrm{MeV}$

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Q.12

Binding energy per nucleon vs mass number curve for nuclei is shown in the Figure. W, X, Y and Z are four nuclei indicated on the curve. The process that would absorb energy is



(A) $Y \rightarrow 2Z$

(B) $W \rightarrow X + Z$

(C) $W \rightarrow 2Y$

(D) $X \rightarrow Y + Z$

Q.13

A proton is bombarded on a stationary Lithium nucleus. As a result of collision two α -particles are produced. The direction of motion of the α particles with the initial direction of motion makes an angle $\cos^{-1}\frac{1}{4}$. If B.E/Nucleon for Li^7 and He^4 are $5.60MeV$ and $7.06MeV$ respectively, then :

(A) Kinetic energy ofstriking proton is $17.28MeV$

(B) Kinetic energy of striking proton is $8.64MeV$

(C) Kinetic energy of striking proton is $4.32MeV$

(D) Kinetic energy of striking proton is $2.16MeV$

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Q.14

The masses of neutron and proton are 1.0087 a.m.u. and 1.0073 a.m.u. respectively. If the neutrons and protons combine to form a helium nucleus (alpha particles) of mass 4.0015 a.m.u the binding energy fo the helium nucleus will be $(1$ a.m.u. $=$ $931\mathrm{MeV})$

(A) $28.4\mathrm{MeV}$

(B) $20.8\mathrm{MeV}$

(C) $27.3\mathrm{MeV}$

(D) $14.2\mathrm{MeV}$

Q.15

The mass of ${}_7\mathrm{N}^{15}$ is $15.00011\mathrm{amu}$, mass of ${}_8\mathrm{O}^{16}$ is $15.99492\mathrm{amu}$ and $\mathrm{m_P} = 1.00783\mathrm{amu}$. Determine binding energy of last proton of ${}_8\mathrm{O}^{16}$.

(A) $2.13\mathrm{MeV}$

(B) $0.13\mathrm{MeV}$

(C) $10\mathrm{MeV}$

(D) $12.13\mathrm{MeV}$

Q.16

A heavy nucleus having mass number 200 gets disintegrated into two small fragments of mass number 80 and 120 . If binding energy per nucleon for parent atom is $6.5\mathrm{MeV}$ and for daughter nuclei is $7\mathrm{MeV}$ and $8\mathrm{MeV}$ respectively, then the energy released in the decay will be -

(A) $200\mathrm{MeV}$

(B) $-220\mathrm{MeV}$

(C) $220\mathrm{MeV}$

(D) $180\mathrm{MeV}$

Q.17

If the binding energy per nucleon in ${}^7_3\mathrm{Li}$ and ${}^4_2$ He nuclei are $5.60\mathrm{MeV}$ and $7.06\mathrm{MeV}$ respectively, then in the reaction

$$\mathrm{p} + {}^7_3\mathrm{Li} \longrightarrow 2{}^4_2\mathrm{He}$$

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energy of proton must be

(A) 28.24MeV

(B) 17.28MeV

(C) 1.46MeV

(D) 39.2MeV

Q.18

If a proton and anti-proton come close to each other and annihilate, how much energy wil be released

(A) $1.5 \times 10^{-10}\ \text{J}$

(B) $3 \times 10^{-10}\ \text{J}$

(C) $4.5 \times 10^{-10}\ \text{J}$

(D) None of these

Q.19

Using the following data
Mass of hydrogen atom = $1.00783\ \ \text{u}$
Mass of neutron = $1.00867\ \ \text{u}$
Mass of nitrogen atom $_7\text{N}^{14} = 14.00307\ \ \text{u}$
The calculated value of the binding energy of the nucleus of the nitrogen atom $_7\text{N}^{14}$ is close to

(A) $56\ \ \text{MeV}$

(B) $98\ \ \text{MeV}$

(C) $104\ \ \text{MeV}$

(D) $112\ \ \text{MeV}$

Q.20

What is the order of magnitude of ratio of the volume of an atom to the volume of its nucleus?

(A) 10^{10}

(B) 10^{15}

(C) 10^{20}

(D) 10^{25}

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Q.1 Answer:

0.0503×10^{-27} kg, 28.3 Mev

Solution:

The symbol $^4_2$ He indicates that the helium neuclus contains $Z = 2$ protons and $N = 4 - 2 = 2$ neutrons. To ohtain the mass defect Δm, we first determine the sum of the individual masses of the separated protons and neutrons. Then, we subtract from this sum the mass of the nucleus.

(a) We find that the sum of the individual masses of the nucleons is

$$\underbrace{2\left(1.6726 \times 10^{-27}\ \text{kg}\right)}_{\text{Two protons}} + \underbrace{2\left(1.6749 \times 10^{-27}\ \text{kg}\right)}_{\text{Two neutrons}}$$
$$= 6.6950 \times 10^{-27}\ \text{kg}$$

This value is greater than the mass of the intact He nucleus and the mass defect is

$$\Delta m = 6.6950 \times 10^{-27}\ \text{kg} - 6.6447 \times 10^{-27}\ \text{kg}$$
$$= 0.0503 \times 10^{-27}\ \text{kg}$$

(b) According to Eq. (iv), the binding energy is given by

$$\Delta E_{\text{BE}} = (\Delta m)c^2 = \left(0.0503 \times 10^{-27}\ \text{kg}\right)\left(3.00 \times 10^8\ \text{m s}^{-1}\right)^2$$
$$= 4.53 \times 10^{-12}\ \text{J}$$

Usually, binding energies are expressed in energy units of electron volt instead of joule $\left(1\text{eV} = 1.60 \times 10^{-19}\ \text{J}\right)$.

$$\therefore \quad \text{Binding energy} = \left(4.53 \times 10^{-12}\ \text{J}\right)\left(\frac{1\text{eV}}{1.60 \times 10^{-19}\ \text{J}}\right)$$
$$= 2.83 \times 10^7 \text{eV} = 28.3\text{MeV}$$

In this result, one million electron volt is denoted by the unit MeV. The value of 28.3MeV is more than two million times greater than the energy required to remove an orbiaal electron from an atom.

Q.2 Answer:

$2.35 \times 10^{17}\ \text{kg m}^{-3}$

Solution:

$$\textbf{Density} = \frac{\text{Mass}}{\text{V olume}}$$
$$\rho = \frac{16\text{amu}}{\frac{4}{3}\pi \text{R}^3}$$
$$\rho = \frac{16 \times 1.67 \times 10^{-27}}{\frac{4}{3}\pi \text{R}^3}$$
$$\rho = 2.35 \times 10^{17}\ \text{kg/m}^3$$

Q.3 Answer:

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4.8

Solution:

$\frac{r_{Cu}}{r_{Al}} = \left(\frac{A_{Cu}}{A_{Al}}\right)^{1/3} = \left(\frac{64}{27}\right)^{1/3}$
$\Rightarrow r_{Cu} = 3.6\left(\frac{4}{3}\right) = 4.8$ **Fermi**

Q.4 Answer:

6×10^{-15} m

Solution:

Radius R **os a nucleus changes with the nucleon number A of the nucleus as** $R = 1.3 \times 10^{-15} \times A^{1/3}$ m

Hence,

$\frac{R_2}{R_1} = \left(\frac{A_2}{A_1}\right)^{1/3} = \left(\frac{128}{16}\right)^{1/3} = (8)^{1/3} = 2$
$\therefore R_2 = 2R_1 = 2\left(3 \times 10^{-15}\right)$ m $= 6 \times 10^{-15}$ m

Q.5 Answer:

19 : 81

Solution:

Let the percentage of B^{10} **atoms be x, then average atomic weight**
$= \frac{10x+11(100-x)}{100} = 10.81 \Rightarrow x = 19$
$\therefore \frac{N_B^{10}}{N_{B^{11}}} = \frac{19}{81}$

Q.6 Answer:

6.0fm.

Solution:

It has been known that a nucleus of mass number A **has radius** $R = R_0 A^{1/3}$
where $R_0 = 1.2 \times 10^{-15}$ m **and** $A =$ **mass number**
In case of ${}^{27}_{13}Al$**, let nuclear radius be** R_1
and for ${}^{125}_{32}Te$**, nuclear radius be** R_2
For ${}^{27}_{13}Al$, $R_1 = R_0(27)^{1/3} = 3R_0$
For ${}^{125}_{32}Te$, $R_2 = R_0(125)^{1/3} = 5R_0$
$\frac{R_2}{R_1} = \frac{5R_0}{3R_0} = \frac{5}{3}R_1 = \frac{5}{3} \times 3.6 = 6$fm

Q.7 Answer:

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Solution:

We use the formula, $R = R_0 A^{1/3}$

This represents relation between atomic mass and radius of the nucleus. For berillium,

$R_1 = R_0(9)^{1/3}$

For germanium, $R_2 = R_0 A^{1/3}$

$\frac{R_1}{R_2} = \frac{(9)^{1/3}}{(A)^{1/3}} \Rightarrow \frac{1}{2} = \frac{(9)^{1/3}}{(A)^{1/3}}$

$\Rightarrow \frac{1}{8} = \frac{9}{A} \Rightarrow A = 8 \times 9 = 72.$

Q.8 Answer:

$1 : 2^{1/3}$

Solution:

$\mathrm{p_1} = \mathrm{p_2}$

$\mathrm{m_1 v_1} = \mathrm{m_2 v_2}$

$2\ \mathrm{m_1} = \mathrm{m_2}$

$2\rho \cdot \frac{4}{3}\pi \mathrm{R_1^3} = \rho \cdot \frac{4}{3}\pi \mathrm{R_2^3}; \frac{\mathrm{R_1^3}}{\mathrm{R_2^3}} = 1 : 2; \mathrm{R_1} : \mathrm{R_2} = 1 : 2^{1/3}$

Q.9 Answer:

$\frac{5}{3}\mathrm{R_{Al}}$

Solution:

As we know, $\mathrm{R} = \mathrm{R_0}(\ \mathrm{A})^{1/3}$

where A **= mass number** $\mathrm{R_{AI}} = \mathrm{R_0}(27)^{1/3} = 3\mathrm{R_0}$

$\mathrm{R_{Te}} = \mathrm{R_0}(125)^{1/3} = 5\mathrm{R_0} = \frac{5}{3}\mathrm{R_{AI}}$

Q.10 Answer:

$1801.7\mathrm{MeV}$

Solution:

Mass defect $\Delta m = [Z m_P + (A - Z) m_n] - M\left(\mathrm{U}^{238}\right)$

$= [92 \times 1.007825(238 - 92) \times 1.008665]$

238.050783

$\Delta m = 1.93421\mathrm{amu}$

$BE = 1.93421 \times 931.5\mathrm{MeV} = 1801.7\mathrm{MeV}$

Q.11 Answer:

$23.8\mathrm{MeV}$

Solution:

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Given, ${}_1\mathrm{H}^2 + {}_1\mathrm{H}^2 \rightarrow {}_2\mathrm{H}^4 + Q$

The total binding energy of the deutrons $= 4 \times 1.15 = 4.60\mathrm{MeV}$

The total binding energy of alpha particle $= 4 \times 7.1 = 28.4\mathrm{MeV}$

The energy released in the process $= 28.4 - 4.60 = 23.8\mathrm{MeV}$

Q.12 Answer:

$W \rightarrow 2Y$

Solution:

$E_W = 120 \times 7.5 = 900$
$E_X = 90 \times 8 = 720$
$E_Y = 60 \times 8.5 = 510$
$E_Z = 30 \times 5 = 150$

$Q_{(1)} = Y - 2Z = 510 - 300 = 210$
$Q_{(2)} = W - X - Z = 900 - 720 - 150 = 30$
$Q_{(3)} = W - 2Y = 900 - 1020 = -120$
$Q_{(4)} = X - Y - Z = 720 - 510 - 150 = 60$

As Q_3 **is negative, energy is absorbed.**

Q.13 Answer:

Kinetic energy ofstriking proton is $17.28\mathrm{MeV}$

Solution:

Q **value of the reaction** $Q = (2 \times 4 \times 7.06 - 7 \times 5.6) = 17.28$ MeV

$K_P + Q = 2K_\alpha$
$\sqrt{2m_P K_P} = 2\sqrt{2m_\alpha K_\alpha}\cos\alpha$
$K_P = K_\alpha$
So $K_P = 17.28\mathrm{MeV}$

Q.14 Answer:

$28.4\mathrm{MeV}$

Solution:

$\mathrm{B.E.} = \Delta \mathrm{mc}^2 = \Delta \times 931\mathrm{MeV}$
$= [2(1.0087 + 1.0073) - 4.0015] \times 931$
$= 28.4\mathrm{MeV}$

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Q.15 Answer:

$12.13\mathrm{MeV}$

Solution:

$\mathrm{M}\left({}_{8}\mathrm{O}^{16}\right)=\mathrm{M}\left({}_{7}\mathrm{N}^{15}\right)+1\ \mathrm{m_P}$

binding energy of last proton $=\mathrm{M}\left(\mathrm{N}^{15}\right)+\mathrm{m_P}-\mathrm{M}\left({}_{1}\mathrm{O}^{16}\right)$

$=15.00011+1.00783-15.99492$

$=0.01302\mathrm{amu}=12.13\mathrm{MeV}$

Q.16 Answer:

$220\mathrm{MeV}$

Solution:

Energy released $=(80\times 7+120\times 8-200\times$
$6.5)$

$=220\mathrm{MeV}$

Q.17 Answer:

$17.28\mathrm{MeV}$

Solution:

Let E **be the energy of proton, then** $E+7\times 5.6=2\times[4\times 7.06]$

$\Rightarrow E=56.48-39.2=17.28\mathrm{MeV}$

Q.18 Answer:

$3\times 10^{-10}\ \mathrm{J}$

Solution:

mass of proton = mass of antiproton

$=1.67\times 10^{-27}\ \mathrm{kg}=\mathrm{laml}$

Energy equivalent to

$1\mathrm{am}=931\mathrm{MeV}$

So energy equivalent to

$2\mathrm{amm}=2\times 931\mathrm{MeV}$

$=1862\times 10^{6}\times 1.6\times 10^{-19}=2.97\times 10^{-10}\ \mathrm{J}$

$=3\times 10^{-10}\ \mathrm{J}$

Q.19 Answer:

$104\ \mathrm{MeV}$

Solution:

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The binding energy of nucleus may be defined as the energy equivalent to the mass defect of the nucleus.

If Δm is mass defect than according to Einstein's mass energy relation.

Binding Energy

$= \Delta mc^2 = Zm_p + A - Zm_n - Mc^2$

$= 7 \times 1.00783 + 7 \times 1.00867 - 14.00307c^2$

or $\text{BE} = 0.1124 \times 931.5\ \text{MeV}$

or $BE \approx 104.6\ \text{MeV}$

Q.20 Answer:

10^{15}

Solution:

Radius of the atom, $R_a = 1\text{Å} = 10^{-10}\text{m}$

Volume of the atom, $V_a = \frac{4}{3}\pi R_a^3 = \frac{4}{3} \times \pi \times 10^{-10^3}\text{m}^3$

Radium of the nucleus, $R_n = 1\text{fermi} = 10^{-15}\text{m}$

Volume of the nucleus. $V_n = \frac{4}{3}\pi R_n^3 = \frac{4}{3} \times \pi \times 10^{-15^3}\text{m}^3$

Their corresponding ratio is $\frac{v_a}{V_n} = \frac{10^{-10^3}}{10^{-15^3}} = \frac{10^{-30}}{10^{-45}} = 10^{15}$

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